Case Study of AIDS Clinical Data via Mixed Model Applications

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May 8, 2026

Abstract

This report explores the applications of four mixed effect models - two linear and two nonlinear - to evaluate the changes between the HIV-1 RNA and CD4 cell count trajectories over the treatment period. These models contained two explanatory variables: time period (t) and CD4 count. All four models considered fixed and random effects, with a comparison of Akaike Information Criterion (AIC) values to determine the best model fit. After best model fit selection, the best linear unbiased predictors (BLUPs) for μ_i and β_i of all four models is explored to determine the effect of fixed and random effects on individual patients and their model predictions. The BLUPs of these coefficients are crucial in understanding the model effects on individual patients, which can assist researchers in evaluating changes in CD4 cell count trajectories over various treatment periods.

Introduction

The goal of the AIDS clinical data set presented is to analyze the effects of inhibitor drugs on HIV type 1 with treatment period and CD4 cell count as the main effects. The response variable is the viral load (V_{ij}) for individual patients (i) and time period (j). Mixed effect models are an appropriate structure for this study since mixed effects account for hierarchical effects between the variables, in this case different patient reactions over points in time.

This report aims to investigate two linear and two nonlinear mixed effect models to determine best model fit based on AIC as well as an estimation of individual trajectories for each model type. The linear mixed effect models attempt to capture linear combinations between time and CD4 count, while the nonlinear mixed effect models will use exponential decay processes as a means of modeling viral load. After best model fit, individual trajectories are estimated to determine the "subject-specific" quantities of viral load by calculating BLUPs of μ_i and β_i, which is key to understanding the effects on individual patients.

About the Data

The data provided contains 361 observations among 6 variables, including patient ID, time, viral load, and CD4 cells. There were originally 53 patients involved in the data, settling to 48 patients after 5 dropped out due to drug intolerance and other factors. Further, a data subset was created for time under 90 days to eliminate chances of contamination. The purpose of the study is modeling the changes between HIV-1 RNA and CD4 cell count trajectories for the population average as well as individual patients. This data was used for the linear and nonlinear mixed model analysis and was run in R version 4.5.2

Linear and Nonlinear Mixed Models

Model Formulas

There are four models that are explored in this paper. The first model is a linear mixed effect model:

$$V_{ij} = \beta_1 + \beta_2 t_{ij} + \beta_3 t_{ij} CD4_{ij} + \upsilon_i + e_{ij}$$

where $\beta = (\beta_1, \beta_2, \beta_3)^T, Var(\upsilon_i) = D = \sigma_\upsilon^2$,Var(e$_i$) $= R_i = \sigma^2 I_{n_i}$.

The random coefficient mixed model is represented as:

$$V_{ij} = \beta_{i1} + \beta_{i2} t_{ij} + \beta_{i3} t_{ij} CD4_{ij} + e_{ij}$$

where $\beta_i = (\beta_{i1}, \beta_{i2}, \beta_{i3})^T$ with $\beta_{ij} = \beta_j + \upsilon_{ij}$ for $j = 1, 2, 3$ and $\upsilon_i = (\upsilon_{i1}, \upsilon_{i2}, \upsilon_{i3})^T$.
where $Var(\upsilon_i) = D_{3\times3}$ and $Var(e_i) = R_i = \sigma^2 I_{n_i}$.

The nonlinear mixed effect model will be written as:

$$V_{ij} = \beta_1 \exp\{-\beta_2 t_{ij}\} + \beta_3 \exp\{-\beta_4 CD4_{ij}\} + \upsilon_i + e_{ij}$$

where $\beta = (\beta_1, \beta_2, \beta_3, \beta_4)^T$

Lastly, the nonlinear random coefficient mixed model is represented as:

$$V_{ij} = \beta_{i1} \exp\{-\beta_{i2} t_{ij}\} + \beta_{i3} \exp\{-\beta_{i4} CD4_{ij}\} + e_{ij}$$

where $\beta_i = (\beta_{i1}, \beta_{i2}, \beta_{i3}, \beta_{i4})^T$ with $\beta_{ij} = \beta_j + \upsilon_{ij}$ for $j = 1, 2, 3, 4$ and $\upsilon_i = (\upsilon_{i1}, \upsilon_{i2}, \upsilon_{i3}, \upsilon_{i4})^T$.
In addition, it is assumed $Var(\upsilon_i) = D_{4\times4}$ and $Var(e_i) = R_i = \sigma^2 I_{n_i}$.

The mixed models all consider random effects on the model, with the nonlinear models incorporating the same starting points for the beta values for consistency ($\beta_1 = 2.5, \beta_2 = 0.05, \beta_3 = 2.5, and \beta_4 = 0.001$). These values were based upon the baseline values, with β_1 of 2.5 falling between the typical baseline value of 2 and 4, the β_2 value based on the exponential time rate, the β_3 value between typical observations for patients, and β_4 based upon the very small CD4 clearance rate from patients.

Model Results

Each model is run in R programming based on the formulas above, accounting for random effects among the β coefficients. The linear models used the maximum likelihood method in R, as this method is used to fit linear mixed-effect models in the lme package. The nonlinear models also used the maximum likelihood method in R but with the nlme package, which uses the same distribution and residual structure as the lme package but for nonlinear models.

After the plots are run in R, the models are plotted with their respective model fits (**Appendix A**). From the visualizations, the two linear models appear to be overfitting with their smooth line jumping from various points. In comparison, the nonlinear models used curves to smooth the data. One nonlinear model appeared to still be overfitting the data (nonlinear mixed effect) while the nonlinear random coefficient mixed effect model appear to have a very smooth predictive line. The β coefficients, variance, and AIC values of the four models were compared:

The results show the linear mixed models and nonlinear mixed models had similar β coefficients and AIC values to their respective model types. Of note, the linear mixed models did not include β_4 as there were only three fixed effects, while the nonlinear mixed models include β_4 to account

	β_1	β_2	β_3	β_4	σ^2	AIC
LME	4.37	-0.02	-0.00	–	0.43	729.20
RCM	4.38	-0.03	-0.00	–	0.37	727.63
NLM	2.42	0.05	2.79	0.001	0.36	602.04
NLRC	2.46	0.07	2.65	0.001	0.15	494.95

Table 1: Model Comparison: Parameters by Model. β_4 is left unfilled for the first two models since they were not included in the equation.

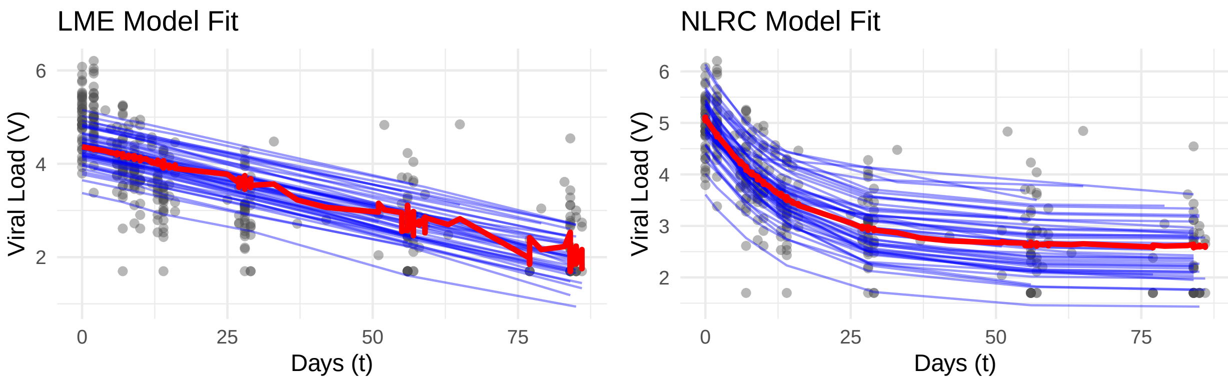

Figure 1: Comparison between LME and NLRC models. The red lines represents the population average, while the blue line represents the subject-specific predictions.

for the exponential rate in CD4 counts. The linear models' coefficients attempt to capture a linear rate of change over time as well as interaction effects between time and CD4, while the nonlinear models' coefficients attempt to capture a decay rate of time and CD4 individually.

From the results, the variance decreased with the nonlinear models, which hints the nonlinear models were a better fit in explaining the residuals in the data. The AIC values also dramatically decreased, which is a sign of better model fit for the predicted values. From the table, the nonlinear random coefficient mixed effect model (NLRC) has the lowest AIC by a large margin, thus this model is the best model based on AIC.

Figure 1 explores one linear model (LME) and one nonlinear model (NLRC). The LME's average line jumps all over the place, hinting at overfitting and lack-of-fit. The NLRC's average line performs much better, with the NLRC model appearing more smooth and a better fit. The LME model uses linear slopes in its prediction, which explains its overfitting and lack-of-fit when the data may be prone to decay rates over time. In contrast, the NLRC model uses curved slopes to account for the random effects in the model, which may explain the more smooth prediction curve in the model. The RCM and NLM model performance can be viewed in **Appendix A**, with the RCM model appearing to have a better fit than its linear model counterpart while the NLM model allows for curved slopes but is prone to overfitting.

Estimation of Individual Trajectories

Subject - Specific Quantities ($\hat{V}_{ij}$)

In addition to best model fit, the estimated individual "subject-specific" quantities were produced for each model, highlighting the accuracy of $\hat{V}_{ij}$ when comparing observed to predicted values at various time periods. From the data given, there are 328 observations when accounting for each patient i at time j. An individual patient's results was viewed and printed to determine the best model predictor (Appendix B). This is only true for one patient, and all patients need to be investigated to determine best predictive model results. The results below show the total count for each model when it was closest to the observed result and its total percentage of times it was closest.

Subject-Specific Prediction Results

Measure	LME	RCM	NLM	NLRC
Count	34	40	89	165
Percentage	10.4%	12.2%	27.1%	50.3%

Table 2: Subject-Specific Predictions results, with the total count the times the specific model was closest to the observed value. The linear mixed effect models performed worse than the nonlinear mixed effect models, with the nonlinear random coefficient mixed effect model having the closest prediction values to the observed values.

From the results, the linear mixed effect models performed worse in prediction than the nonlinear models. This is not unexpected since the linear versions create linear slopes in their estimations while the nonlinear models create curved slopes via random effects, which allows the nonlinear models to capture more variability. Overall, the two nonlinear models contained 77.4% of the best subject-specific prediction results. Further, the NLRC version produced the highest percentage of accuracy, with 50.3% of the 165 of the 328 observations having the closest prediction by the NLRC model.

Estimates of β

The estimates of β represent the population average of the effects and can be viewed in the table 1 results, with β_1, β_2, and β_3 values printed for the two linear mixed models and all four β coefficients printed for the two nonlinear mixed models. This is because β_1 is the intercept value, β_2 is the value for t, and β_3 is the value for the interaction term between t and CD4. β_3 is near zero for both linear mixed effect models, meaning they have hardly any effect in the model. For the two other models, β_4 is a fixed effect for the nonlinear mixed effect model but was an individual patient effect on the nonlinear random coefficient mixed effect model. When viewing the β coefficients, the intercept values are very similar when comparing the linear models to one another as well as the nonlinear models. β_3 appears to have a large impact on the nonlinear models, indicating an impact of CD4 count on viral loads in the nonlinear models.

BLUPs of μ_i and β_i

This section reviews how to acquire the best linear unbiased predictors (BLUPs) of μ_i and β_i, with μ_i representing the random effects of each subject i and β_i representing subject-specific coefficient for each subject i. In the context of μ_i, this is the difference between the population average of β

and the performance of each individual i. For BLUPs, it is assumed $\mu_i \sim$ N(0, σ_u^2). Below are the results for displaying the random effect values related to the random coefficient mixed model:

Random Effects of RCM Model

Patient ID	Intercept	t	t:CD4
11542	-0.32	-0.02	5.03e-05
11960	-1.18	0.02	-3.74e-05
250009	-0.30	0.01	-2.49e-05
250405	-0.92	-0.02	-3.50e-05
250422	-0.38	-0.02	4.74e-05

Table 3: Random Effects of RCM model. Positive and negative values represent the relationship between the random effect and population average.

For the most part, a positive random effect value represents a higher value than the population average, while a negative random effect values represents a lower value than the population average. In the case of the RCM model, a positive intercept means a higher baseline viral load average and a negative intercept means a lower baseline viral load average. For t, a positive value highlights a slower decay rate in the time slope while a negative value highlights a faster decay rate in the time slope, as seen in the visualizations of the model. For $t : CD4$ effect, a positive μ_i means a patient has stronger sensitivity to CD4 effects while a negative μ_i indicates a patient has a weaker sensitivity to CD4 effects. Each model have varying random effects, as seen in the model equations at the beginning of the report. From the tables of the first 5 patients for each individual model (**Appendix C**), it appears the random effects of the NLRC model had a stronger impact on CD4 effect (β_3), which may be the reason for better model fit.

The BLUPs of β_i are the subject-specific coefficients for each individual. This is adding the random effect coefficient to the fixed effect coefficient ($\beta_i = \mu_i + \beta$). The β_i coefficients assist in prediction accuracy for each individual patient, as μ_i accounts for potential noise and adds to the group average (β). Below is a summary table for the β_i coefficients of the linear mixed model:

Beta Coefficients of LME Model

Patient ID	Intercept	t	t:CD4
11542	**3.97**	-0.02	-2.74e-05
11960	**3.37**	-0.02	-2.74e-05
250009	**4.18**	-0.02	-2.74e-05
250405	**3.65**	-0.02	-2.74e-05
250422	**4.16**	-0.02	-2.74e-05

Table 4: Beta Coefficients of LME model. Only the Intercept value is changed since μ_i only accounted for the Intercept values.

As can be seen from the table, only the intercept values differ between the patients. This is because the random effects for the LME model only impacted the intercept, so the time slope variable and CD4 interaction term will be unchanged in the β_i coefficient values, and are therefore the same for all patients. In comparison, the β_i coefficients for the NLRC were the following for the first 5 patients:

From the NLRC β_i results, β_1 (Intercept) and β_3 (CD4 scaling) differ since the random effects μ_i were associated with these values whereas β_2 (decay rate) and β_4 (CD4 rate) are the fixed effects

Beta Coefficients of LME Model

Patient ID	β_1	β_2	β_3	β_4
11542	**2.36**	0.07	**2.25**	6.94e-05
11960	**2.18**	0.07	**1.46**	6.94e-05
250009	**2.20**	0.07	**2.67**	6.94e-05
250405	**2.03**	0.07	**2.00**	6.94e-05
250422	**3.37**	0.07	**1.84**	6.94e-05

Table 5: Beta Coefficients of NLRC model. β_1 and β_3 are different due to μ_i affecting these values.

and will not vary between patients. So, the NLRC model considers a different starting point and difference in scaling relative to CD4 effects over time for each patient, which explains why this model performs better than the other models in both model fit and prediction.

Conclusion

This report explores the impact of each mixed effect model on the AIDS Clinical data, successfully fitting a model for the two linear and two nonlinear examples. From the AIC values and visualizations presented, the linear mixed effect models performed poorly and do not appear to fit the data well, while the nonlinear versions are able to fit the data better. In the end, the nonlinear random coefficient mixed effect model was the best choice, with an AIC value of 494.95 and a smooth fit line in its visualization. From the equations of the models, this may be due to the NLRC model considering both fixed and random effects on CD4 and the decay rate of time, which would help smooth nonlinear data.

After selecting the best model via AIC, the random effects (μ_i) and subject-specific coefficients (β_i) for each individual were produced for each model. The random effects and subject-specific coefficients were then interpreted, with intercept values indicating an individual's performance to the baseline, time slopes related to rate of decay, and CD4 effect for scaling purposes. In the end, the report was able to successfully create BLUPs of both μ_i and β_i for individual patients, which can greatly assist researchers in the assessment of immunological response of an antiviral regimen.

Appendix

Appendix A

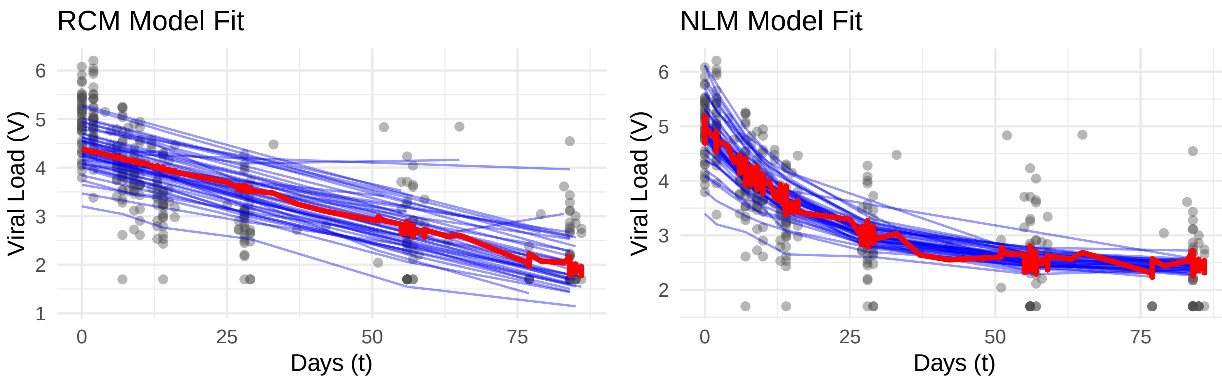

Figure 2: Comparison between RCM and NLM models. The red lines represents the population average, while the blue line represents the subject-specific predictions.

Appendix B

Subject-Specific Predictions for Patient ID 11542

Time	Observed	LME	RCM	NLM	NLRC
0	4.36	3.97	4.07	**4.37**	4.57
7	3.53	3.79	3.82	3.82	**3.66**
16	2.98	3.56	3.51	3.35	**2.99**
29	2.64	3.25	3.02	2.99	**2.53**

Table 6: Subject-Specific Predictions fo patient 11542, comparing the observed value to the predicted value of each model. The nonlinear random coefficient model was consistently the closest to the observed value for this patient.

Appendix C

Random Effects of LME Model

Patient ID	Intercept
11542	-0.399
11960	-0.992
250009	-0.191
250405	-0.719
250422	-0.404

Table 7: Random Effects of LME model, with values related to the intercept.

Random Effects of NLM Model

Patient ID	β_1
11542	-0.530
11960	-1.478
250009	-0.453
250405	-1.183
250422	-0.090

Table 8: Random Effects of NLM model, with values related to the intercept.

Random Effects of NLRC Model

Patient ID	β_1	β_3
11542	-0.100	-0.404
11960	-1.478	-1.192
250009	-0.453	0.019
250405	-1.183	-0.649
250422	-0.090	-0.812

Table 9: Random Effects of NLRC model, containing values related to the intercept and CD4 effects.

R Code

```r
library(nlme)
library(ggplot2)
data <- read.table("ACTG315.dat", header=TRUE)
colnames(data) <- c("No1", "No2", "patid", "t", "V", "CD4")
set.seed(123)
#only include first 3 months
data_sub <- subset(data, t <= 90)

#Linear mixed effect model
model_lme <- lme(V ~ t + t:CD4,
                 random = ~ 1 | patid,
                 data = data_sub,
                 method = "ML")
summary(model_lme) #AIC = 729.20

#Random coefficient mixed effect model
model_rcm <- lme(V ~ t + t:CD4,
                 random = ~ t + t:CD4 | patid,
                 data = data_sub,
                 method = "ML",
                 control = lmeControl(opt = "optim"))
summary(model_rcm) #AIC = 727.63


#Nonlinear mixed effect
model_nlm <- nlme(V ~ b1*exp(-b2*t) + b3*exp(-b4*CD4),
                  data = data_sub,
                  fixed = b1 + b2 + b3 + b4 ~ 1,
                  random = b1 ~ 1 | patid,
                  start = c(b1=2.5, b2=0.05, b3=2.5, b4=0.001))
summary(model_nlm) #AIC = 602.04

#Nonlinear random coefficient mixed effect model --look at initial start values
model_nlrc <- nlme(V ~ b1*exp(-b2*t) + b3*exp(-b4*CD4),
                  data = data_sub,
                  fixed = b1 + b2 + b3 + b4 ~ 1,
                  random = b1 + b3 ~ 1 | patid,
                  start = c(b1=2.5, b2=0.05, b3=2.5, b4=0.001))
summary(model_nlrc) #AIC = 494.95

#Compare AIC values
aic_values <- AIC(model_lme, model_rcm, model_nlm, model_nlrc)
print(aic_values) #NLRC with clear lowest AIC

##Model Visualizations##

#Predictions for LME
#Level 1 = Subject-Specific -- blue line
#Level 0 = Population Average -- red line
data_sub$pred_indiv_lme <- predict(model_lme, level = 1)
data_sub$pred_pop_lme   <- predict(model_lme, level = 0)
#Plot LME
lme_plot <- ggplot(data_sub, aes(x = t, y = V, group = patid)) +
  # Raw data points
```

```r
  geom_point(alpha = 0.4, color = "grey30") +
  #Subject-specific predictions (Individualized fit)
  geom_line(aes(y = pred_indiv_lme), color = "blue", alpha = 0.4) +
  #Population-level prediction (The true average of your model)
  geom_line(aes(y = pred_pop_lme, group = NULL), color = "red", linewidth = 1.2) +
  theme_minimal() +
  labs(
    title = "LME Model Fit",
    subtitle = "Blue: Subject-Specific | Red: Population Average",
    y = "Viral Load (V)",
    x = "Days (t)"
  )

#Predictions for RCM
data_sub$pred_indiv_rcm <- predict(model_rcm, level = 1)
data_sub$pred_pop_rcm   <- predict(model_rcm, level = 0)

#RCM plot
rcm_plot <- ggplot(data_sub, aes(x = t, y = V, group = patid)) +
  #Raw data points
  geom_point(alpha = 0.4, color = "grey30") +
  #Subject-specific predictions
  geom_line(aes(y = pred_indiv_rcm), color = "blue", alpha = 0.4) +
  #Population-level prediction
  geom_line(aes(y = pred_pop_rcm, group = NULL), color = "red", linewidth = 1.2) +
  theme_minimal() +
  labs(
    title = "RCM Model Fit",
    subtitle = "Blue: Subject-Specific | Red: Population Average",
    y = "Viral Load (V)",
    x = "Days (t)"
  )

#Predictions for NLM
data_sub$pred_indiv_nlm <- predict(model_nlm, level = 1)
data_sub$pred_pop_nlm   <- predict(model_nlm, level = 0)

#NLM plot
nlm_plot <- ggplot(data_sub, aes(x = t, y = V, group = patid)) +
  # Raw data points
  geom_point(alpha = 0.4, color = "grey30") +
  # Subject-specific predictions
  geom_line(aes(y = pred_indiv_nlm), color = "blue", alpha = 0.4) +
  # Population-level prediction
  geom_line(aes(y = pred_pop_nlm, group = NULL), color = "red", linewidth = 1.2) +
  theme_minimal() +
  labs(
    title = "NLM Model Fit",
    subtitle = "Blue: Subject-Specific | Red: Population Average",
    y = "Viral Load (V)",
    x = "Days (t)"
  )

#Predictions for NLRC
data_sub$pred_indiv_nlrc <- predict(model_nlrc, level = 1)
data_sub$pred_pop_nlrc   <- predict(model_nlrc, level = 0)
```

```
#NLRC plot
nlrc_plot <- ggplot(data_sub, aes(x = t, y = V, group = patid)) +
  # Raw data points
  geom_point(alpha = 0.4, color = "grey30") +
  # Subject-specific predictions
  geom_line(aes(y = pred_indiv_nlrc), color = "blue", alpha = 0.4) +
  # Population-level prediction
  geom_line(aes(y = pred_pop_nlrc, group = NULL), color = "red", linewidth = 1.2) +
  theme_minimal() +
  labs(
    title = "NLRC Model Fit",
    subtitle = "Blue: Subject-Specific | Red: Population Average",
    y = "Viral Load (V)",
    x = "Days (t)"
  )


# Initialize a list of models
model_list <- list("LME" = model_lme, "RCM" = model_rcm, "NLM" = model_nlm,
                    "NLRC" = model_nlrc)

# Define the parameter names as they should appear in the table
param_names <- c("Beta 1", "Beta 2", "Beta 3", "Beta 4", "Sigma^2", "AIC")

# Create an empty matrix to store results
results_matrix <- matrix(NA, nrow = length(param_names), ncol = 4)
colnames(results_matrix) <- c("LME", "RCM", "NLM", "NLRC")
rownames(results_matrix) <- param_names

# Loop through models and manually extract values
for (i in 1:length(model_list)) {
  mod <- model_list[[i]]
  f_eff <- fixef(mod)

  # Fill Fixed Effects
  results_matrix[1, i] <- round(f_eff[1], 4) # Beta 1
  results_matrix[2, i] <- round(f_eff[2], 4) # Beta 2
  results_matrix[3, i] <- round(f_eff[3], 4) # Beta 3

  # Models C and D have a 4th beta
  if (length(f_eff) >= 4) {
    results_matrix[4, i] <- round(f_eff[4], 4)
  }

  # Extract Sigma^2 (Residual Variance)
  results_matrix[5, i] <- round(mod$sigma^2, 4)

  # Extract AIC
  results_matrix[6, i] <- round(AIC(mod), 2)
}

flipped_matrix <- t(results_matrix)

# Convert to data frame for easier handling
final_table_flipped <- as.data.frame(flipped_matrix)
```

```r

##Extract subject-specific predictions##
for (m_name in names(model_list)) {
  mod <- model_list[[m_name]]

  cat("\n--- Subject-Specific Predictions for Model", m_name, "---\n")

  # Obtain predicted values (V-hat_ij) for each observation
  # these use the individual-level coefficients
  subject_predictions <- predict(mod, level = 1)

  # Combine with original data for a clear output
  output_data <- data.frame(
    patid = data_sub$patid,
    Time  = data_sub$t,
    Observed_V = data_sub$V,
    Predicted_V = subject_predictions
  )

  print(head(output_data))
}

#Combines all results into one data frame
all_results <- data.frame(
  Observed = data_sub$V,
  Pred_LME    = predict(model_lme, level = 1),
  Pred_RCM    = predict(model_rcm, level = 1),
  Pred_NLM    = predict(model_nlm, level = 1),
  Pred_NLRC    = predict(model_nlrc, level = 1)
)

#Absolute error of each model
all_results$Error_LME <- abs(all_results$Observed - all_results$Pred_LME)
all_results$Error_RCM <- abs(all_results$Observed - all_results$Pred_RCM)
all_results$Error_NLM <- abs(all_results$Observed - all_results$Pred_NLM)
all_results$Error_NLRC <- abs(all_results$Observed - all_results$Pred_NLRC)

#Identify which models have the smallest errors
error_cols <- all_results[, c("Error_LME", "Error_RCM", "Error_NLM", "Error_NLRC")]
all_results$Best_Model_Index <- apply(error_cols, 1, which.min)

#Index model names
model_names <- c("Model LME", "Model RCM", "Model NLM", "Model NLRC")
all_results$Best_Model_Name <- model_names[all_results$Best_Model_Index]

#Prediction counts based on smallest errors
prediction_counts <- table(all_results$Best_Model_Name)
print(prediction_counts)

#Percentages from the prediction counts
prediction_percentages <- prop.table(prediction_counts) * 100
print(prediction_percentages)

#Obtain beta coefficients for each patient
fixef_summary <- list(
  LME  = fixef(model_lme),
```

```r
  RCM  = fixef(model_rcm),
  NLM  = fixef(model_nlm),
  NLRC = fixef(model_nlrc)
)

fixef_summary

#Extract BLUPs for Linear Mixed Model
mu_lme <- ranef(model_lme)
head(mu_lme)

# Extract BLUPs for the Random Coefficient Model
mu_rcm <- ranef(model_rcm)
head(mu_rcm)

# Extract BLUPs for the Nonlinear Mixed Model
mu_nlm <- ranef(model_nlm)
head(mu_nlm)

# Extract BLUPs for the Nonlinear Random Coefficient Model
mu_nlrc <- ranef(model_nlrc)
head(mu_nlrc)

#Obtain beta_i for each model
beta_i_lme  <- coef(model_lme)
beta_i_rcm  <- coef(model_rcm)
beta_i_nlm  <- coef(model_nlm)
beta_i_nlrc <- coef(model_nlrc)

#View results for beta_i for each model
head(beta_i_lme)
head(beta_i_rcm)
head(beta_i_nlm)
head(beta_i_nlrc)
```

Listing 1: R Source Code